Pressure Field Generated by a Free-Falling Body in Earth's Atmosphere: A Conceptual Extension of Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law of Gravity)

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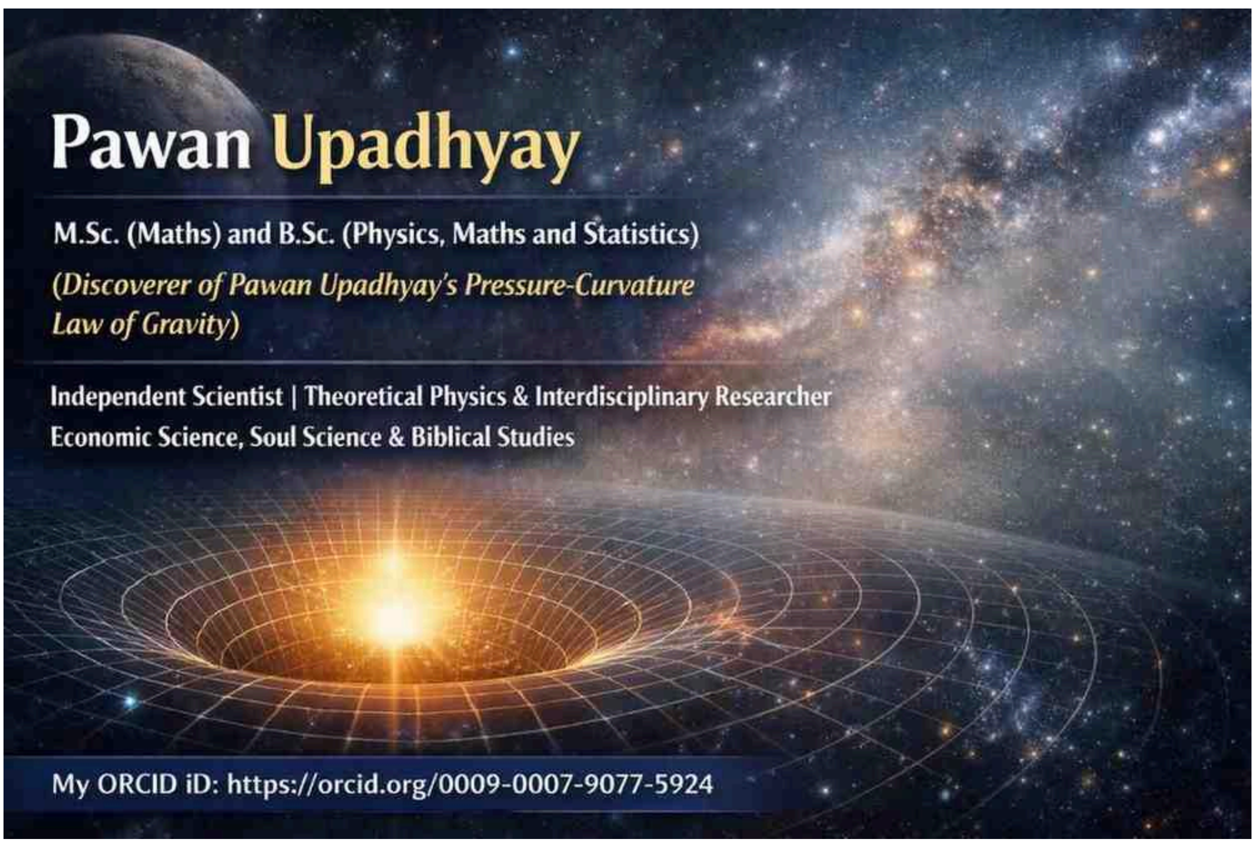

Abstract

When a body falls freely through Earth's atmosphere, it interacts continuously with the surrounding air, producing localized pressure distributions that give rise to aerodynamic drag. In conventional fluid mechanics, these pressure fields are understood as consequences of fluid flow and are not regarded as sources of gravity. Within **Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law)**, however, pressure is interpreted as a fundamental physical quantity associated with spacetime curvature. This paper proposes a conceptual extension of the PPC framework by distinguishing Earth's large-scale gravitational pressure field from the localized atmospheric pressure field generated by a moving body. The proposed interpretation suggests that these pressure fields coexist and influence motion through different physical mechanisms. This work is presented as a theoretical hypothesis requiring future mathematical formulation and experimental verification.

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1. Introduction

A body released from an aircraft undergoes free fall under Earth's gravitational influence while simultaneously moving through the atmosphere. As it descends, the body continuously interacts with the surrounding air, producing changes in pressure that result in aerodynamic drag.

Classical fluid mechanics explains these effects entirely through pressure distributions generated by the body's motion through the fluid. General Relativity, on the other hand, describes free fall as inertial motion along geodesics determined by spacetime curvature, with air resistance acting as a non-gravitational force.

Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law) interprets gravitational curvature as arising from pressure associated with mass–energy density. This paper explores whether the localized atmospheric pressure generated during free fall can be conceptually incorporated within the broader gravitational pressure framework proposed by PPC.

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2. Localized Atmospheric Pressure Field During Free Fall

As a body moves through Earth's atmosphere, it compresses the air ahead of its motion while displacing the surrounding fluid. This interaction produces a localized pressure distribution around the body.

In classical fluid dynamics, this pressure distribution generates aerodynamic drag, which opposes the body's motion. The magnitude of the drag depends on factors such as body shape, velocity, atmospheric density, and the resulting pressure gradients around the object.

The localized atmospheric pressure field is therefore a dynamic consequence of fluid interaction rather than a direct manifestation of Earth's gravitational field.

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3. Interpretation Within the PPC Framework

Within the PPC Law, pressure is regarded as a fundamental quantity associated with spacetime curvature generated by mass–energy density. From this perspective, a free-falling body moving through Earth's atmosphere may be interpreted as existing within two distinct pressure environments.

3.1 Earth's Gravitational Pressure Field

Earth's mass–energy generates the large-scale gravitational pressure field that determines the curvature of spacetime and governs the body's geodesic motion.

3.2 Localized Atmospheric Pressure Field

Simultaneously, the body's motion through air generates a localized atmospheric pressure field responsible for aerodynamic drag. This pressure field arises from fluid interaction and modifies the body's trajectory relative to ideal geodesic free fall.

According to this conceptual interpretation, the localized atmospheric pressure field exists within Earth's background gravitational pressure field without replacing or canceling it.

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4. Combined Interpretation

Within this proposed extension of the PPC framework, the motion of a free-falling body may be understood as resulting from two simultaneously existing pressure fields:

- **Earth's gravitational pressure field**, which governs the underlying geodesic motion through spacetime.

- **The localized atmospheric pressure field**, which produces aerodynamic drag through fluid interaction.

These pressure fields operate through different physical mechanisms. The gravitational pressure field determines the background geometry of motion, while the localized atmospheric pressure field introduces non-gravitational resistance arising from interaction with the surrounding fluid.

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5. Future Research

Future work should investigate whether localized pressure fields generated by moving bodies can be incorporated mathematically into the PPC framework. In particular, it remains to be explored whether dynamic pressure distributions can be related consistently to the pressure-based interpretation of spacetime curvature without altering the established predictions of General Relativity.

Potential research directions include:

- Developing a mathematical description of localized pressure fields within the PPC framework.
- Investigating interactions between gravitational pressure and dynamic atmospheric pressure.
- Determining whether measurable effects beyond conventional fluid dynamics arise under specific physical conditions.
- Examining consistency with the stress-energy tensor and the existing equations retained within the PPC framework.

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6. Conclusion

A body in free fall through Earth's atmosphere simultaneously experiences Earth's gravitational environment and localized pressure distributions generated by its motion through air. Classical physics attributes these localized pressure fields to fluid dynamics, where they produce aerodynamic drag.

Within the Pressure–Curvature Law of Gravity, this paper proposes a conceptual interpretation in which the localized atmospheric pressure field coexists with Earth's gravitational pressure field. The former modifies motion through fluid interaction, while the latter governs the underlying spacetime geometry responsible for free fall.

This proposal is intended as a conceptual extension of the PPC framework. It does not constitute an established prediction of the theory and requires rigorous mathematical formulation, consistency with the underlying gravitational equations, and experimental verification before it can be regarded as a physical result.

Acknowledgment

The author acknowledges the foundational contributions of classical fluid mechanics and Albert Einstein's General Relativity, whose established physical principles provide the basis for comparison with the conceptual interpretation proposed within Pawan Upadhyay's Pressure–Curvature Law of Gravity.

One-line takeaway:
A free-falling body moving through Earth's atmosphere exists simultaneously within Earth's gravitational pressure field and a localized atmospheric pressure field generated by its motion; this coexistence is proposed here as a conceptual extension of the Pressure–Curvature Law of Gravity and remains a hypothesis requiring mathematical and experimental validation.

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6. Explanation of the sentence of 'Conclusion' :

In the sentence:

"The former modifies motion through fluid interaction, while the latter governs the underlying spacetime geometry responsible for free fall."

The former refers to **the localized atmospheric pressure field**.

This means:

- As a body falls through the atmosphere, it compresses the air in front of it and creates pressure differences around its surface.
- These pressure differences produce **aerodynamic drag**.
- Drag **changes the body's motion** by reducing its acceleration below the ideal free-fall value, decreasing its speed relative to a vacuum, and eventually causing it to reach terminal velocity.

Thus, **"modifies motion"** means that the atmospheric pressure field changes **how the body moves through the air**, not the underlying gravitational field.

"The localized atmospheric pressure field modifies the body's observed motion by producing aerodynamic drag through fluid interaction, whereas Earth's gravitational pressure field governs the underlying geodesic motion responsible for free fall."

This wording clearly distinguishes the two roles:

- **Earth's gravitational pressure field** → determines the underlying free-fall trajectory (in the PPC interpretation).
- **Localized atmospheric pressure field** → introduces aerodynamic drag that alters the body's actual motion through the atmosphere.